Analyzing SVM vs QSVM: Mathematical Foundations and Experimental Results

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Abstract

Quantum Support Vector Machines (QSVM) are a promising application of quantum computing in machine learning. This research proposes a novel hybrid QSVM model that integrates gate-based quantum computation for kernel construction and adiabatic quantum computation for optimization. The proposed approach seeks to address the limitations of standalone quantum paradigms while achieving higher accuracy and computational efficiency. This paper presents an in-depth analysis of the hybrid model and its advantages over classical SVM and individual quantum approaches.

Keywords

Quantum Computing, Machine Learning, Support Vector Machines, Quantum Optimization, Quantum Kernel.

1. Introduction

Quantum computing, as an evolving sub-discipline within computer science, amalgamates various research fields, including physics, mathematics, and computer science, among

many others. The course of technological advancement has divided the field into two distinct approaches: gate-based quantum computation and adiabatic quantum computation. Although these approaches are theoretically equivalent, their practical utilization diverges significantly. Gate-based quantum computation is studied for a more generalpurpose application, whereas adiabatic computation is primarily employed for quantum annealing, an optimization process aimed at determining the minimum of a cost function. Quantum machine learning (QML) stands as a heavily studied sub-discipline within the framework of quantum computation, it being popular within both the gate-based and adiabatic paradigms. This article focuses on the Support Vector Machine (SVM) model and describes how these two divergent quantum computing approaches seek to implement it. We propose a novel approach to model this problem, combining both technologies. Our model is based on the observation that the two quantum approaches for the Quantum Support Vector Machine (QSVM) focus on two different parts of the classical approach. Gate-based computation aims to leverage quantum properties to discover a useful kernel for the high-dimensional Hilbert space. On the other hand, the annealing model focuses on optimization that comes after computing the kernel matrix. These are two separate but complementary components of the Support Vector Machine model. The combination of these approaches, if correctly applied, provides advantages over the classical method, as well as each approach on its own. The following sections perform an in-depth analysis of the current advancements in Quantum Support Vector Machines. It begins with an exposition on classical SVM, then delves into an investigation of the distinct approaches employed in computing QSVM.

1.1. Classical Support Vector Machine :

It is a common supervised algorithm to classify through support vector machines . We use the kernel trick to put our data in a higher dimension where it become linearly separable by applying a function .
Basic idea of support vector machines :

- Optimal hyperplane for linearly separable patterns
- Extend to patterns that are not linearly separable by transformations of original

data to map into new space – the Kernel function .

Support vectors are the data points that lie closest to the hyperplane . In 2 dimensions , we can separate by a line but in higher dimensions we need hyperplanes .

We can draw many boundaries, but one can argue that the best is one that maximizes

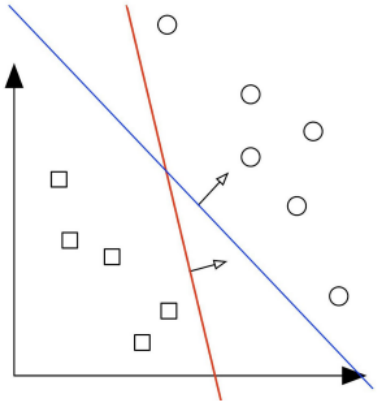

Figure 1: 2-D Case

the margin between the closest data point to the boundary .

M is the training data points of the form

$$(\vec{x}_j, y_j) : \vec{x}_j \in \mathbb{R}^N, \; y_j = \pm 1$$

where $y_j = +1$ or -1.

Let $\vec{w}$ represent the norm to the decision boundary. The margin is given by two parallel hyperplanes, separated by a distance $\frac{2}{\|\vec{w}\|}$ with no data points inside the margin.

The decision boundary of the two hyperplanes:

$$\vec{w} \cdot \vec{x} - b = +1$$

$$\vec{w} \cdot \vec{x} - b = -1$$

The training data is correctly classified if

$$\vec{w} \cdot \vec{x}_i - b \geq +1 \quad \text{if } y_i = +1$$

$$\vec{w} \cdot \vec{x}_i - b \leq -1 \quad \text{if } y_i = -1$$

So for all i,

$$y_i(\vec{w} \cdot \vec{x}_i - b) \geq +1 \quad \text{for all } i$$

Slack Variables for Soft Margin:

The constraint with slack variables:

$$y_i(\mathbf{w} \cdot \mathbf{x}_i - b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

where ξ_i allows for misclassification or margin violations.

Final Objective Function (Primal Form):

The optimization problem:

$$\min_{\mathbf{w},b,\xi} \quad \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n} \xi_i$$

Subject to:

$$y_i(\mathbf{w} \cdot \mathbf{x}_i - b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \text{for all } i.$$

Dual Formulation of SVM

Since the problem is a quadratic program and the strong duality holds, we can take the dual.

The Lagrangian: The Lagrangian is formulated to incorporate constraints into the objective function.

$$L(\vec{w}, b, \vec{\alpha}) = \frac{1}{2}\|\vec{w}\|^2 + \sum_{i=1}^{n} \alpha_i \left(1 - y_i(\vec{w} \cdot \vec{x}_i - b)\right)$$

Primal Optimization:

$$p^* = \min_{\vec{w},b} \max_{\alpha_i \geq 0} L(\vec{w}, b, \vec{\alpha})$$

Stationary Conditions: The partial derivatives of the Lagrangian with respect to w

and b are set to 0 to find stationary points .

$$\frac{\partial L(\vec{w}, b, \vec{\alpha})}{\partial \vec{w}} = \vec{w} - \sum_{i=1}^{n} \alpha_i y_i \vec{x}_i = 0 \quad \Rightarrow \quad \vec{w} = \sum_{i=1}^{n} \alpha_i y_i \vec{x}_i$$
$$\frac{\partial L(\vec{w}, b, \vec{\alpha})}{\partial b} = \sum_{i=1}^{n} \alpha_i y_i = 0$$

The Lagrangian:

$$L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\mathbf{w}\|^2 + \sum_{i=1}^{n} \alpha_i \left(1 - y_i(\mathbf{w} \cdot \mathbf{x}_i - b)\right)$$

Primal Problem:

$$p^* = \min_{\vec{w}, b} \max_{\alpha_i \geq 0} L(\vec{w}, b, \vec{\alpha})$$

Dual Problem: Using strong duality (since SVM is a quadratic programming problem), the roles of minimization and maximization are reversed. The dual problem is:

$$d^* = \max_{\alpha_i \geq 0} \min_{\vec{w}, b} L(\vec{w}, b, \vec{\alpha})$$

Simplified Dual Formulation: After eliminating $\vec{w}$ and b, the dual problem becomes:

$$d^* = \max_{\alpha_i \geq 0} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j (\mathbf{x}_i \cdot \mathbf{x}_j)$$

$$\sum_{i=1}^{n} \alpha_i y_i = 0$$

- $\mathbf{x}_i \cdot \mathbf{x}_j$: Represents the inner product of feature vectors, which can be replaced by a kernel function for non-linear decision boundaries.
- α_i: Represents the Lagrange multipliers, with non-zero values corresponding to the support vectors.

The dual optimization problem in SVMs is defined as:

$$L(\vec{\alpha}) = \sum_{j=1}^{M} y_j \alpha_j - \frac{1}{2} \sum_{j,k=1}^{M} \alpha_j \alpha_k K_{jk},$$

where:

- α_j are the Lagrange multipliers.
- $y_j \in \{-1, 1\}$ are the class labels.
- K_{jk} is the kernel function, $K_{jk} = \vec{x}_j \cdot \vec{x}_k$ for a linear kernel.

The optimization is subject to the following constraints:

$$\sum_{j=1}^{M} \alpha_j = 0, \quad y_j \alpha_j \geq 0 \quad \forall j.$$

Hyperplane Parameters

After solving for α, the hyperplane parameters are recovered as:

$$\vec{w} = \sum_{j=1}^{M} \alpha_j y_j \vec{x}_j,$$

and the bias b is computed as:

$$b = y_i - \vec{w} \cdot \vec{x}_i,$$

where $\vec{x}_i$ is a support vector.

Support Vectors

Only a few α_j are non-zero, corresponding to the support vectors lying on the hyperplane margins.

Kernel Representation and Classification in SVMs

The kernel matrix K_{jk} is defined as:

$$K_{jk} = k(\vec{x}_j, \vec{x}_k) = \vec{x}_j \cdot \vec{x}_k,$$

where $k(\vec{x}_j, \vec{x}_k)$ is the kernel function. For a linear kernel:

$$k(\vec{x}_j, \vec{x}_k) = \vec{x}_j \cdot \vec{x}_k.$$

The hyperplane parameters are recovered, and only a few of the α_j are non-zero. These correspond to the support vectors that define the margins of the hyperplane.

Binary Classifier

The decision function for classifying a new vector $\vec{x}$ is:

$$y(\vec{x}) = \operatorname{sgn}\left(\sum_{j=1}^{M} \alpha_j k(\vec{x}_j, \vec{x}) + b\right),$$

where:

- $\operatorname{sgn}(\cdot)$: The sign function, returning $+1$ or -1.
- b: The bias term, computed from the support vectors.

The result is a binary classifier that predicts $y = +1$ or $y = -1$ for a new input vector $\vec{x}$.

2. Problem Formulation

The detection of credit card fraud remains a significant challenge due to the ever-increasing volume of transactions and the highly imbalanced nature of fraud datasets. Fraudulent transactions represent only a small fraction of the overall dataset, making the development of reliable and efficient detection systems crucial. It should be noted that this domain has indeed shown promise by the use of classical machine learning methods, which include Support Vector Machines (SVM); however, when handling complex high-dimensional patterns, as those in finance, the performance of SVM might become limited. Quantum machine learning, specifically the Quantum Support Vector Machine (QSVM), offers a novel approach to address these challenges. By leveraging quantum computing's ability to process information in a high-dimensional Hilbert space, QSVMs can potentially provide advantages in terms of feature representation, model complexity, and

computational efficiency. However, the practical applicability and performance of QSVMs in real-world scenarios, such as credit card fraud detection, remain underexplored.
This study aims to investigate and compare the effectiveness of classical SVM and QSVM models for credit card fraud detection. The objectives of this work are as follows:

- **Develop and Implement Models:** To construct and train both a classical SVM and a QSVM using the same credit card transaction dataset to ensure a fair comparison.
- **Performance Evaluation:** Evaluating both models using standardized metrics such as accuracy, precision, recall, F1-score, and AUC-ROC, with a focus on detecting fraudulent transactions.
- **Comparison of Computational Efficiency:** Comparing the computational cost of QSVM with the classical SVM by determining the training time and inference time to gauge if QSVMs are viable in real-world applications.

Through this comparative analysis, the study aims to determine whether quantum-enhanced machine learning approaches, such as QSVM, can outperform classical methods in detecting credit card fraud and provide insights into their potential advantages and limitations in tackling real-world classification problems.

3. Methodology

3.1. Data Preprocessing and validation

Data preprocessing and validation are critical steps in ensuring the reliability and robustness of the quantum support vector machine (QSVM) for credit card fraud detection. During preprocessing, the raw credit card transaction dataset is cleaned, normalized, and balanced to address issues such as missing values, outliers, and class imbalance. Class balancing is particularly important given the highly imbalanced nature of fraud detection datasets, where fraudulent transactions are rare compared to legitimate ones. Validation strategies, such as k-fold cross-validation, are employed to split the dataset into training and testing subsets, ensuring that the QSVM model is trained on diverse data while its

performance is evaluated on unseen data. This step mitigates the risks of overfitting and underfitting, thereby enhancing the model's generalization capability.

3.2. Data Embedding in QSVM

In quantum machine learning, embedding classical data into the quantum space is a pivotal step, as it translates real-world data into a format suitable for quantum computation. For this project, angle encoding is employed, which maps each feature of the classical data into the quantum state space by encoding it as the rotation angle of a quantum qubit. This approach allows the QSVM to exploit quantum parallelism and entanglement, thereby capturing complex relationships in the data that are challenging for classical models. By leveraging angle encoding, the quantum kernel can process high-dimensional data efficiently, enabling the QSVM to distinguish between fraudulent and legitimate transactions with enhanced accuracy.

3.3. Evaluation matrix

The performance of the QSVM is assessed using a comprehensive set of evaluation metrics tailored to the challenges of fraud detection. Metrics such as accuracy, precision, recall, and F1-score are calculated to provide a detailed understanding of the model's predictive capabilities. Given the imbalanced nature of the dataset, special emphasis is placed on recall (sensitivity) and the F1-score, as they measure the ability of the model to correctly identify fraudulent transactions while maintaining a balance between precision and recall. Additionally, the area under the Receiver Operating Characteristic (ROC) curve (AUC-ROC) is used to evaluate the model's overall discrimination power. This multi-metric approach ensures that the QSVM's effectiveness is comprehensively evaluated and benchmarked against classical methods.

4. Results and Discussion

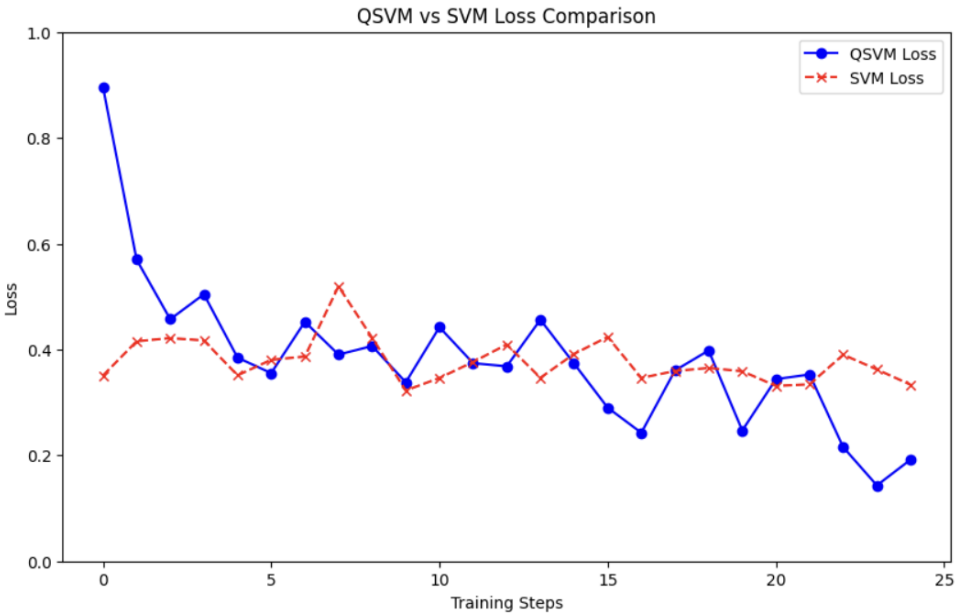

Figure 2: QSVM vs. SVM Loss Comparison

QSVM vs. SVM Loss Comparison :

The graph compares the loss trends of QSVM and SVM over training steps. QSVM has a steeper initial decline in loss, reflecting faster convergence during early training stages. However, its loss shows more variance throughout training compared to SVM. In contrast, SVM has a more stable and consistent loss trend but converges slower in the initial stages.

QSVM Advantage: Quantum computing might help in faster convergence at the very beginning, perhaps because it scans complex feature space more efficiently. This would prove helpful in better detection of fraud patterns much early in training.

SVM Stability: The stability in the loss of SVMs shows robust optimization, which can be expected considering its established approach of training by kernel.

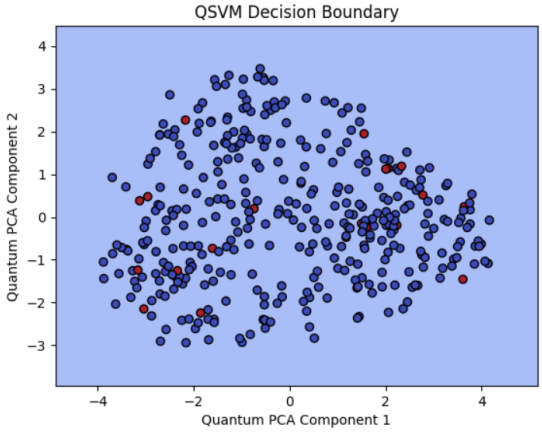

Figure 3: QSVM decision boundary

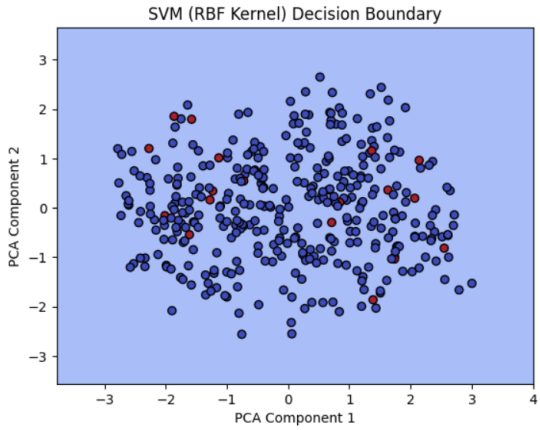

Figure 4: RBF kernel decision boundary

Decision Boundary of QSVM :

Observation: QSVM boundary correctly distinguishes points in quantum transformed feature space; Classification regions indicated with the background as blue. Blue color dot and red are indications for fraud, not a fraud data points.

Insight: The quantum kernel allows mapping data into high-dimensional spaces, where separability of complex patterns, such as those in fraud detection, is more feasible. This indicates QSVM's potential to handle non-linear patterns inherent in credit card fraud datasets.

SVM (RBF Kernel) Decision Boundary :

Observation: The decision boundary created by SVM with an RBF kernel displays good separability between classes. Like QSVM, it distinguishes between fraudulent and non-fraudulent transactions but the boundary is slightly less complex than QSVM.

Insight: The RBF kernel effectively captures non-linear relationships that are essential for fraud detection. Though the boundary is less complex than QSVM, the model is reliable as it possesses good generalization.

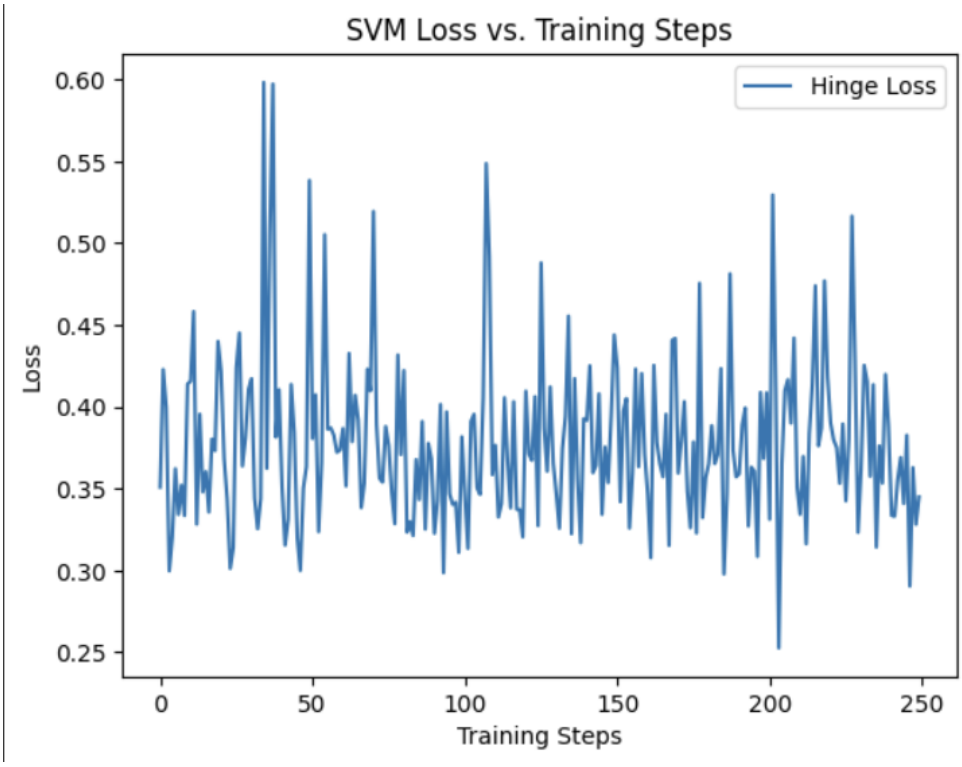

Figure 5: SVM Loss vs. Training Steps

SVM Loss vs. Training Steps :

Notice: The SVM loss plot exhibits oscillations in training, especially at the beginning steps, before stabilizing. The loss is highly oscillatory but converges to a steady state with minimal oscillations.

Insight: The oscillations indicate that the optimizer is searching the parameter space to converge to an optimal solution. SVM's eventual stabilization shows convergence to a solution, thus confirming its stability over time even though it started noisily.

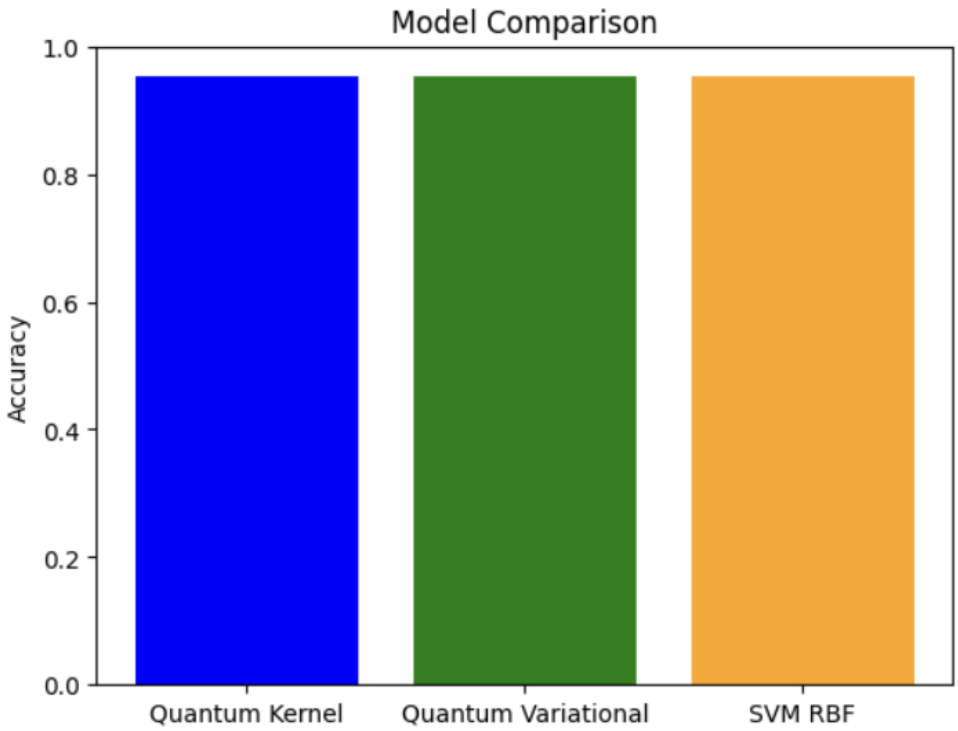

Figure 6: Model Comparison (Accuracy)

Model Comparison (Accuracy) :

Observation: The bar chart compares the accuracy of three models:

Quantum Kernel Model (Blue)

Quantum Variational Model (Green)

SVM with RBF Kernel (Orange)

All models are equally accurate, close to or at 100%, meaning that all of them are good classifiers for fraudulent and non-fraudulent transactions.

Insight:Quantum Models: Quantum kernel and variational models are at par with SVM, indicating that they are able to be as good as classical approaches but in the exploration of quantum feature spaces.

SVM Performance: The classical SVM shows competitive accuracy, which justifies its applicability and efficiency in fraud detection tasks.

General Conclusion:

The results suggest that QSVM has some benefits, such as faster convergence and potentially more complex decision boundaries. However, SVM provides similar accuracy and stable training, which may make it preferable in cases of resource constraint or when the quantum hardware is not accessible. The analysis here shows that QSVM is promising

for high-dimensional fraud detection tasks, but its advantages must be weighed against the practical limitations and computational costs of quantum computing.

References

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